NSE Intelligent Investor

Platform Technical Reference

ET AI Hackathon 2026 · Problem Statement 6

4 AI Modules	LSTM + HMM + LLM	NSE Universe
M1 · M2 · M3 · M4	Multi-Paradigm ML	Nifty 50 Coverage

Contents

1 · User Workflow

The entire platform runs inside a single HTML dashboard connected to a FastAPI backend. The workflow follows 6 sequential steps — from startup to crisis management.

Step 1 — Initialization & Data Integrity

- User opens the HTML file or visits the deployed URL.
- Backend API triggers the startup lifespan event.
- Module 4 (Regime Engine) loads and aligns all 4 data sources.
- SHA-256 hash is generated for each dataset — simulating a blockchain ledger for data provenance.
- A Disclaimer Modal appears: platform is algorithmic, not a registered financial advisor.

Step 2 — Regime Awareness (Module 4)

- Dashboard header displays current market regime: Bull / Bear / Sideways.
- HMM confidence score shown (e.g., '85% Bull confidence').
- If extreme conditions detected → Circuit Breaker overrides HMM → forces Capital Preservation mode.
- UI colour scheme shifts to red to signal danger state.

Step 3 — Opportunity Radar (Module 1)

- User navigates to Opportunity Radar tab.
- System fetches ranked signals from 3 data streams: Corporate Filings, Insider Trades, Bulk Deals.
- LSTM Anomaly Detector filters noise → produces composite score (0–100) per stock.
- User clicks a stock → LLM generates plain-English explanation of why it is flagged.
- Action buttons: 'Optimize with this stock' or 'Check Patterns'.

Step 4 — Chart Pattern Validation (Module 3)

- User searches flagged stock in Chart Patterns module.
- System scans recent OHLCV data and detects 8 classic technical patterns.
- LSTM Continuation Scorer outputs probability (0–100%) that pattern will succeed.
- Historical Backtester shows stock-specific win-rate and sample count.
- Each pattern card shows: entry price, stop-loss, target price, risk-reward ratio.

Step 5 — Portfolio Optimization (Module 2)

- User adds selected stocks to 'My Portfolio' and opens Portfolio Optimizer.
- Selects risk tolerance, investment horizon (5 / 10 / 15 / 30 days), and objective.
- LSTM Predictor forecasts annualized returns for each stock.
- MPT Optimizer computes Efficient Frontier → outputs optimal weight per stock.
- Full risk dashboard generated: Sharpe, Sortino, CVaR, Max Drawdown, Alpha, Beta.
- Rebalancing Proposal generated: exact BUY / SELL / HOLD actions with ■ amounts.

Step 6 — Black Swan Crisis Management

- Background Circuit Breaker detects a >3% daily Nifty drop or VIX spike.
- Critical alert appears on dashboard with real-time portfolio damage estimate.
- Stop-loss positions are flagged if breach thresholds are crossed.
- User queries Loss Recovery Engine: 'How long until my portfolio recovers?'
- System returns: median recovery days + probability of full recovery within 120 days.
- Algorithmic recommendation: Hold (if recovery < 10 days) or Exit and Redeploy.

Example Scenario — Rahul's Full Session

Step	Action	Platform Output
1 · Open App	Rahul launches the dashboard	Regime header shows: Bull Market — 85% Confidence
2 · Discovery	Checks Opportunity Radar	INFY flagged: Score 92 — heavy insider buying + positive filing
3 · Validation	Opens Chart Patterns for INFY	Inverse H&S; detected · LSTM score 78% · Historical win-rate 65%
4 · Optimize	Runs optimizer: INFY + TCS + RELIANCE · ■1,00,000	Weights: INFY 45%, TCS 30%, RELIANCE 25% · Sharpe maximized
5 · Execute	Places trades on broker per Rebalancing Proposal	Exact ■ BUY/SELL/HOLD plan provided
6 · Crisis	Market crashes -4% two weeks later	Circuit Breaker fires · Recovery Engine: INFY median recovery = 8 days → HOLD

2 · Model Training Logic & Assumptions

Architecture Overview

Module	Model Type	Paradigm	Purpose
M1 · Opportunity Radar	LSTM Autoencoder	Unsupervised	Anomaly detection on bulk/insider/filing data
M2 · Portfolio Optimizer	LSTM + MPT (SLSQP)	Supervised + Convex Optimisation	Predict returns → optimize portfolio weights
M3 · Chart Patterns	LSTM Classifier	Supervised Binary Classification	Score probability of pattern continuation
M4 · Regime Engine	Gaussian HMM	Unsupervised (Baum-Welch)	Detect hidden market state: Bull/Bear/Sideways

Module 1 — Opportunity Radar (LSTM Autoencoder)

Architecture

`LSTM(128) → Dropout(0.2) → LSTM(64) → Dense(1)`

Inputs (60-day rolling window, 5 features)

- India VIX (fear index)
- Daily price return of the stock
- Volume ratio (today vs 20-day average)
- Insider activity frequency (trades per day)
- Bulk deal frequency (deals per day)

Training Paradigm

- Unsupervised reconstruction — model learns to compress and rebuild normal 60-day sequences.
- Loss function: Mean Squared Error (MSE) between input and reconstructed output.
- Normal market → low reconstruction error. Anomalous market → high reconstruction error.
- High MSE = anomaly score = multiplier that boosts the base signal score.

Key Assumptions

- Smart money moves (insiders / bulk deals) create subtle statistical anomalies in price and volume.
- Anomaly = Opportunity — unusual activity around fundamental news predicts near-term price momentum.

Module 2 — Portfolio Optimizer (LSTM + MPT)

LSTM Return Predictor Architecture

`LSTM(128) → Dropout(0.3) → LSTM(64) → Dense(1)`

Separate model trained per prediction horizon: 5d, 10d, 15d, 30d

Inputs (60-day window per stock)

- OHLCV data (Open, High, Low, Close, Volume)
- Technical indicators: RSI, MACD
- Macro features: India VIX, sector return

MPT Optimization

- Solver: scipy.optimize.minimize with SLSQP (Sequential Least SQuares Programming).

- Objective: Maximize Sharpe Ratio = (w^T * mu - Rf) / sqrt(w^T * Sigma * w).
- Constraints:
  - Weights sum to exactly 1.0 (fully invested).
  - Individual weight bounds: 0.5% minimum to 30% maximum per stock.
- mu (expected returns) = LSTM predictions. This replaces backward-looking historical means.
- Sigma (covariance matrix) = 252-day rolling historical covariance.

Key Assumptions

- LSTMs capture non-linear dependencies in price history better than simple historical averages.
- 252-day rolling covariance is a reliable proxy for near-term future risk.
- LSTM alpha generation + MPT variance minimization outperforms equal-weighting.

Module 3 — Chart Pattern Intelligence (LSTM Classifier)

Pattern Detection (Rule-Based)

- 5-bar rolling window scans OHLCV data for swing highs and swing lows.
- Detects 8 patterns: Bullish Breakout, Bearish Breakdown, Head & Shoulders, Inverse H&S;, Double Top, Double Bottom, Support Bounce, Resistance Rejection.

LSTM Scorer Architecture

```
LSTM(64) → Dropout(0.3) → Dense(1, sigmoid)
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Inputs (30-day window, 8 engineered features)

- RSI (Relative Strength Index)
- MACD signal line
- Volume ratio (today vs average)
- Normalised ATR (Average True Range — volatility proxy)

Training Paradigm

- Supervised binary classification: label = 1 if pattern succeeded within 10 days, 0 if it failed.
- Output: probability (0–1) of pattern continuation.
- Back-tester computes stock-specific historical win-rate and sample count — not generic textbook stats.

Key Assumptions

- Market psychology repeats itself in geometric price formations.
- Pattern alone is insufficient — 30-day technical context determines fake-out vs genuine continuation.

Module 4 — QuantVault Regime Engine (Gaussian HMM)

Architecture

- Model: Gaussian Hidden Markov Model via hmmlearn library.
- Hidden states: 3 (Bear, Sideways, Bull).
- Training: Unsupervised maximum likelihood estimation using Baum-Welch algorithm.
- State mapping: deterministic sort by mean return — State 0 (lowest) = Bear, State 2 (highest) = Bull.

Input Features (4 daily features)

Feature	Description	Why It Matters
nifty_return	Daily Nifty 50 return	Primary return signal
rolling_vol_10d	10-day rolling return volatility	Regime volatility signature

Feature	Description	Why It Matters
gsec_yield	10-Year G-Sec bond yield	Macro interest rate context
sentiment_score	NLP-derived daily market sentiment	Behavioural/mood signal

Observed Output from Live Run

Regime	Days	Mean Daily Return	Mean Volatility	Mean Yield	Mean Sentiment
■ Bear	238	-0.007%	0.918%	7.23%	-0.010
■ Sideways	180	+0.099%	1.230%	6.63%	+0.051
■ Bull	297	+0.125%	0.536%	7.20%	+0.073

Key Assumptions

- Financial markets operate in distinct hidden regimes with different mean-variance profiles.
- Macroeconomic indicators (yields) and sentiment help HMM delineate regimes without lagging like moving averages.

3 · Technical & Mathematical Analysis

3.1 · CAPM Metrics

Metric	Formula	Interpretation
Alpha (a)	a = Rp - [Rf + b x (Rm - Rf)]	+2.0% means the AI portfolio beat its mathematically expected return by 2%
Beta (b)	b = Cov(Rp, Rm) / Var(Rm)	b < 1 = defensive (drops less in crash) | b > 1 = aggressive | b = 1 = tracks market

- Rf = 10-Year G-Sec yield (India's risk-free rate proxy).
- Rm = Nifty 50 daily returns.
- Alpha is the value added purely by the algorithm's intelligence — independent of market movement.

3.2 · Risk Management Metrics

Metric	Formula	Platform Use
Sharpe Ratio	Sharpe = (Rp - Rf) / sp	Optimizer maximizes this during Bull regime
Sortino Ratio	Sortino = (Rp - Rf) / sd (sd = downside std dev only)	Penalizes only negative volatility — fairer measure for investors
CVaR (95%)	Mean of worst 5% daily returns in historical window	Optimizer switches to minimize CVaR during Bear regime
Max Drawdown	MDD = (Trough - Peak) / Peak	Worst peak-to-trough loss before new high — capital preservation check
Effective N	1 / HHI where HHI = sum(wi^2)	True diversification level — e.g., Effective N = 4.8 means ~5 stocks drive returns

- sp = annualized standard deviation of all daily returns.
- sd = standard deviation of only negative (downside) daily returns.
- CVaR answers: 'On my worst 5% of days, what is my average loss?' — stronger than standard VaR.

3.3 · MPT Convex Optimization

- Objective function — Maximize Sharpe:
  - Maximize: (w^T * mu - Rf) / sqrt(w^T * Sigma * w)
  - w = weight vector | mu = LSTM-predicted returns (not historical mean) | Sigma = 252-day covariance matrix
- Constraints:
  - Sum of all weights = 1.0 (fully invested, no cash position)
  - 0.005 <= wi <= 0.30 (each stock: min 0.5%, max 30%)
- Solver: scipy.optimize.minimize with SLSQP algorithm.
- Key innovation: mu uses LSTM-predicted returns instead of backward-looking historical means.

3.4 · Machine Learning Mathematics

LSTM Anomaly Score (Module 1)

- MSE = (1/n) * sum((xi - x_hat_i)^2) where xi = original, x_hat_i = reconstructed.
- High MSE → unusual sequence → high anomaly score → signal multiplier applied.
- Low MSE → normal market → signal suppressed.

Gaussian HMM — Key Mathematics (Module 4)

- Latent states: Z_t in {Bear, Sideways, Bull} — unobservable, inferred from data.
- Transition matrix A (3x3):
  - A_ij = P(Z_t = j | Z_(t-1) = i)
  - Example: A[Bear][Bull] = probability of market moving from Bear to Bull tomorrow.
- Observed live transition matrix:

From \ To	Bear	Sideways	Bull
Bear	91.5%	1.7%	6.8%
Sideways	2.2%	97.8%	0.0%
Bull	5.4%	0.0%	94.6%

All diagonal values > 90% → regimes are highly stable → once in Bull, very likely to stay Bull.

3.5 · Circuit Breaker Mathematical Logic

Trigger	Threshold	System Response
Daily Nifty drop	> 3%	WARNING state — heightened monitoring
Daily Nifty drop	> 5%	CRITICAL state — optimizer suppresses all BUY signals
10-day rolling vol	>= 2.5%	Circuit Breaker trips — defensive rebalance mode
India VIX	> 25	Elevated fear — HMM confidence reduced
India VIX	> 35	Extreme fear — HMM forced to 100% Bear classification

4 · Black Swan Handling — Circuit Breaker & Recovery Engine

Standard ML models (LSTMs, HMMs) are trained on historical data and can lag or fail during unprecedented crash events. Two dedicated sub-systems override them.

4.1 · Circuit Breaker Engine

Real-Time Triggers

Trigger	Threshold	State Entered
Daily Nifty drop	> 3%	WARNING
Daily Nifty drop	> 5%	CRITICAL
10-day rolling vol	>= 2.5%	Circuit Breaker TRIPPED
India VIX	> 25	Elevated Fear
India VIX	> 35	Extreme Fear — forced 100% Bear

System Override Actions (when trigger fires)

- HMM Override: Standard Regime Engine output is overridden regardless of math.
- UI immediately shifts to forced Bear / Capital Preservation Mode.
- Portfolio Optimizer suppresses all new BUY recommendations.
- Optimizer shifts objective from 'Maximize Sharpe' to 'Minimize CVaR'.
- Dashboard UI colours shift to red — visual danger alert.

4.2 · Real-Time Portfolio Damage Assessment

- Proxy Valuation: Calculates estimated real-time loss for every holding based on the day's drop percentage.
- Stop-Loss Audit: Cross-references current proxy prices against user-defined stop-loss levels.
- Breach Alert: Immediately flags positions that have crossed their exit threshold.
- Critical Alert Example: 'CRITICAL: Portfolio down >10%. Immediate defensive rebalance recommended.'

4.3 · Loss Recovery Engine

What It Solves

- Answers the investor's most urgent question: 'How long until I get my money back?'
- Prevents irrational panic selling with data-backed evidence.

How It Works

- On a massive drawdown, engine searches entire historical NSE dataset.
- Finds every prior instance where that specific stock dropped by a similar magnitude.
- Calculates and displays:
  - Median Recovery Days — average trading days to return to pre-drop price.
  - Probability of Recovery — % of historical instances that recovered within 120 trading days.

Algorithmic Recommendation Rules

Condition	Recommendation	Reasoning
Median recovery > 6 months	EXIT and redeploy capital	Historically too slow to recover
Median recovery < 10 days	HOLD — do not panic sell	Fast historical recovery pattern
Recovery probability < 50%	EXIT — unfavourable odds	More instances failed than succeeded

Condition	Recommendation	Reasoning
Recovery probability > 75%	HOLD with confidence	Strong historical recovery record

Example: 'INFY typically recovers from a -4% drop in 8 trading days (historical median). Recovery probability: 82% within 120 days. Recommendation: HOLD.'

5 · Novelty & Uniqueness

5.1 · Multi-Paradigm AI Architecture

- 4 distinct AI/ML paradigms in one platform — no existing retail tool combines all of these:
  - ■ Unsupervised LSTMs (Autoencoders) — anomaly detection on bulk deals and insider trades.
  - ■ Supervised LSTMs — predict chart pattern continuation probability.
  - ■ Hidden Markov Models (HMMs) — probabilistic macro regime detection.
  - ■ Large Language Models (Claude/GPT) — translate model outputs into plain-English explanations.

5.2 · Regime-Aware Portfolio Optimization

- Traditional MPT assumes static market conditions — this system does not.
- HMM continuously monitors hidden market state and feeds it into the optimizer.
- Regime-dependent behavior:
  - ■ Bull regime → Maximize Sharpe ratio, higher equity exposure.
  - ■ Bear regime → Switch objective to Minimize CVaR, propose defensive rebalancing.
  - ■ This acts as a dynamic, automatic hedge without user intervention.

5.3 · Black Swan Loss Recovery Engine

- No existing retail platform answers: 'How long will it take to get my money back?'
- Scans decades of historical NSE data upon a crash event.
- Outputs median recovery time (days) + probability of full recovery — asset-specific, not generic.
- Prevents panic selling with hard mathematical evidence from the same asset's own history.

5.4 · Blockchain-Inspired Data Immutability

- Problem: 'Garbage in, garbage out' — corrupted data = corrupted signals.
- All 4 critical data streams (OHLCV, yields, sentiment, returns) are SHA-256 hashed on ingestion.
- Hashes shown on the Regime Radar dashboard with 'VERIFIED' badges.
- Same data always produces the same hash — any tampering produces a completely different hash.
- Provides institutional-level auditability accessible to retail investors.

5.5 · Extreme Transparency via UI/UX

- All quant logic wrapped in a glassmorphic single-page Vanilla JS frontend — no Jupyter notebook required.
- Complex institutional metrics exposed through intuitive visuals:
  - ■ Efficient Frontier curve — interactive risk-return plot.
  - ■ 3x3 HMM Transition Probability Matrix — rendered as a colour-coded grid.
  - ■ Rebalancing Drift table — exact ■ BUY / SELL / HOLD per stock.
- LLM converts every AI score into plain-English rationale — accessible to first-time investors.
- Win-rates are per-stock, per-pattern (not generic textbook statistics).

6 · Competitive Comparison

Comparison of NSE Intelligent Investor against the leading retail investment platforms in India.

Feature / Capability	NSE Intelligent Investor	TradingView	Tickertape / Screener	Smallcase
Target Audience	Advanced Retail / Quant Enthusiasts	Technical Traders	Fundamental Investors	Passive / Thematic
Core Value Proposition	End-to-end AI Quant pipeline	Advanced Charting & Social	Stock Screening & Financials	Ready-made baskets
Technical Analysis	**Yes — algorithmic detection of 8 patterns**	**Yes — industry-leading manual charts**	Basic only	None
Deep Learning Predictions	**Yes — LSTM for returns & patterns**	No — historical indicators only	No	No
Market Regime Detection	**Yes — HMM Bull/Bear/Sideways**	No	No	No
Portfolio Optimization	**Yes — MPT + LSTM hybrid + Efficient Frontier**	No	No	No — pre-defined weights only
Black Swan Failsafes	**Yes — Circuit Breaker + VIX monitor**	Alerts only	No	No
Loss Recovery Estimates	**Yes — historical time-to-breakeven engine**	No	No	No
Automated Rebalancing	**Yes — exact BUY/SELL/HOLD with ■ amounts**	No	No	Periodic updates by creators
AI Plain-English Explanations	**Yes — LLM integration**	No — community scripts	No	No
Data Integrity Verification	**Yes — SHA-256 ledger hashing**	No	No	No
Pricing / Access	Open Source / Local Deployment	Freemium (Pro = expensive)	Freemium	Transaction + subscription fees

Summary of Competitive Advantage

- TradingView — best manual charting tool, but purely passive. Requires the user to form their own strategy.
- Tickertape / Screener — best for fundamental data aggregation, but no predictive layer.
- Smallcase — provides strategies, but they are opaque, static, and not personalized.
- NSE Intelligent Investor is the only platform that:
  - Acts as an active algorithmic partner — not just a data display tool.
  - Applies institutional-grade ML (LSTMs + HMMs) to predict probabilities.
  - Constructs mathematically optimal, personalized portfolios.
  - Deploys hard-coded capital preservation during extreme market crashes.
  - Explains every output in plain English — accessible to all investor types.